

MAE 1001 · FALL 2026

Python “Power Chords”

Introduction to Mechanical and Aerospace Engineering

Course website: <https://gwu-mae1001.github.io/fall2026/>

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Course site

Photo: Kartik Bulusu

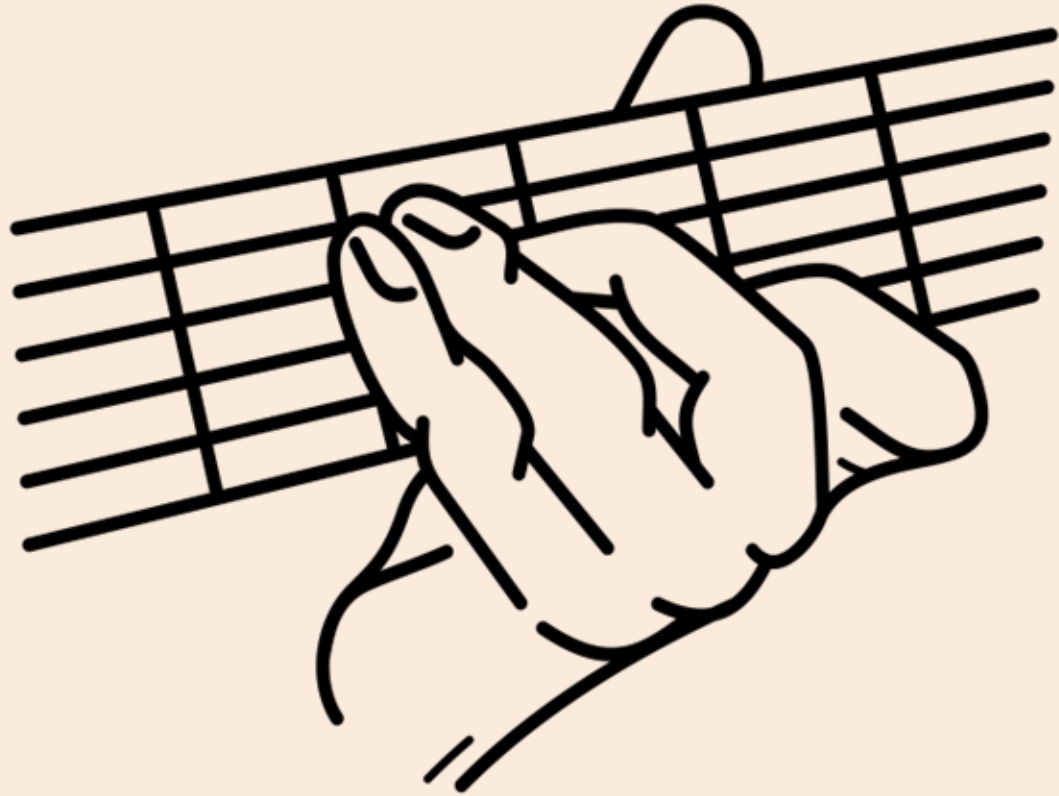
A THOUGHT BEFORE WE START



“Study hard what interests you the most in the most undisciplined, irreverent and original manner possible.”

— Richard Feynmann





What is a Power Chord?

- A power chord consists of just two notes: the root and the fifth.
- Commonly used in rock music for its strong, clear sound.

Representing Power Chords in Python

- Musical intervals can be modeled with **arrays/lists**, making it easy to calculate the root and fifth for any note.



Let's hear it: power chords in Python

```
import numpy as np
import sounddevice as sd

def play_power_chord(root_freq, duration=1.0, fs=44100):
    fifth_freq = root_freq * 2**(7/12)
    t = np.linspace(0, duration, int(fs*duration), endpoint=False)
    tone = np.sin(2*np.pi*root_freq*t) + np.sin(2*np.pi*fifth_freq*t)
    tone = tone / np.max(np.abs(tone)) # Normalize
    sd.play(tone, fs)
    sd.wait()

play_power_chord(82.41)
play_power_chord(110)
play_power_chord(146.83)
play_power_chord(196)
play_power_chord(246.94)
play_power_chord(329.63)
```

What you'll hear

Each call plays a root + fifth on one open guitar string (standard tuning).

E2 82.41 Hz

A2 110 Hz

D3 146.83 Hz

G3 196 Hz

B3 246.94 Hz

E4 329.63 Hz

fifth = root × 2^(7/12)



Thonny-editor or integrated development environment (IDE)

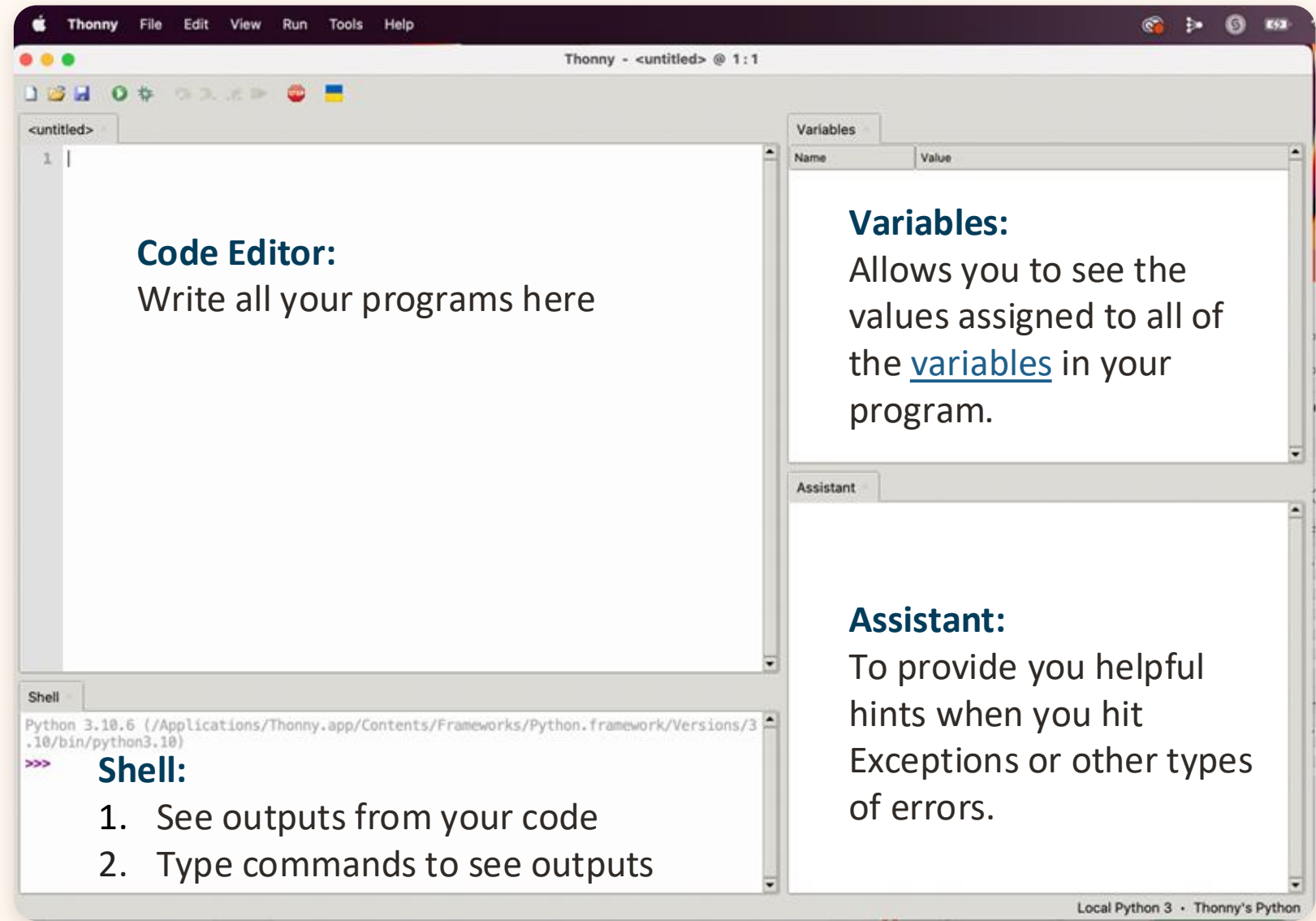
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GET STARTED WITH RASPBERRY PI

Source: <https://www.raspberrypi.org/help/>

What are the other options?

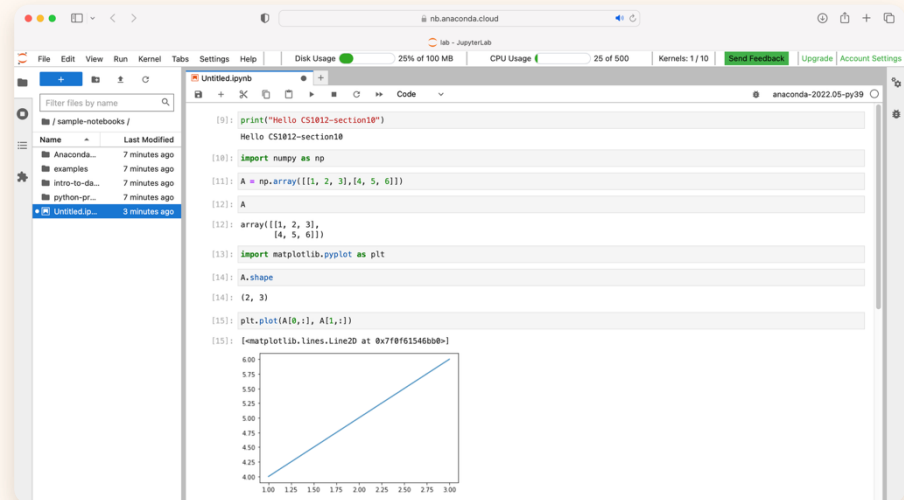
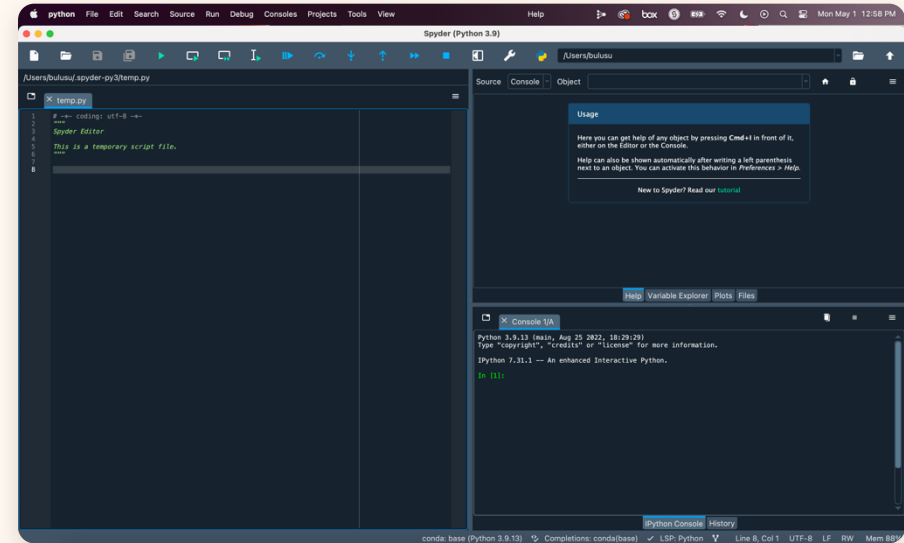
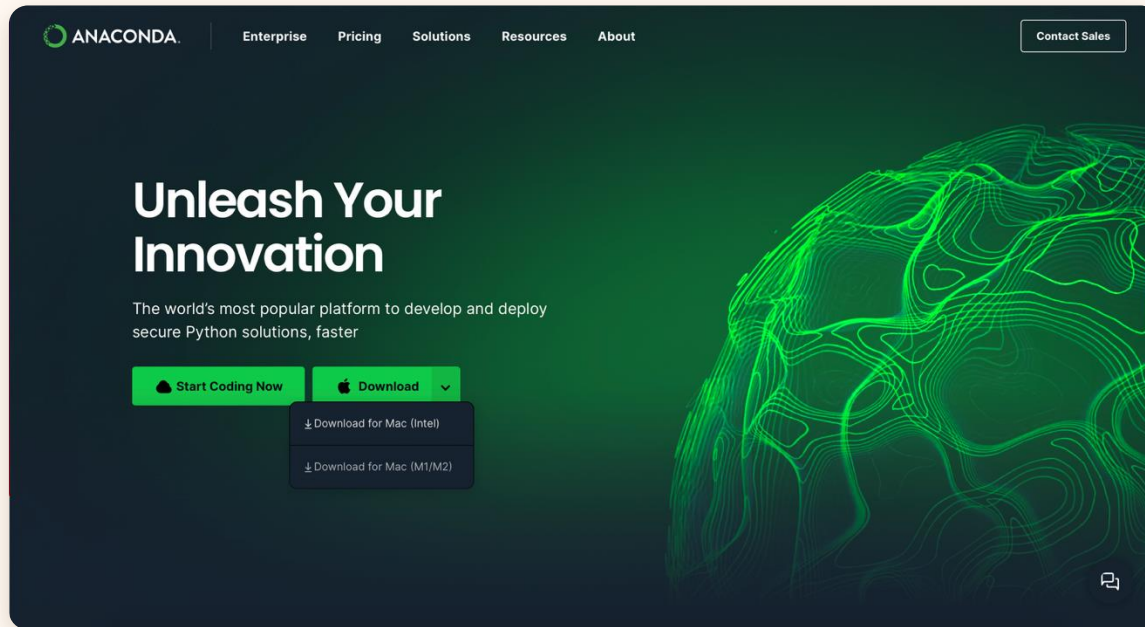


Anaconda python – IDEs:

Spyder

Jupyter Notebook

Anaconda cloud



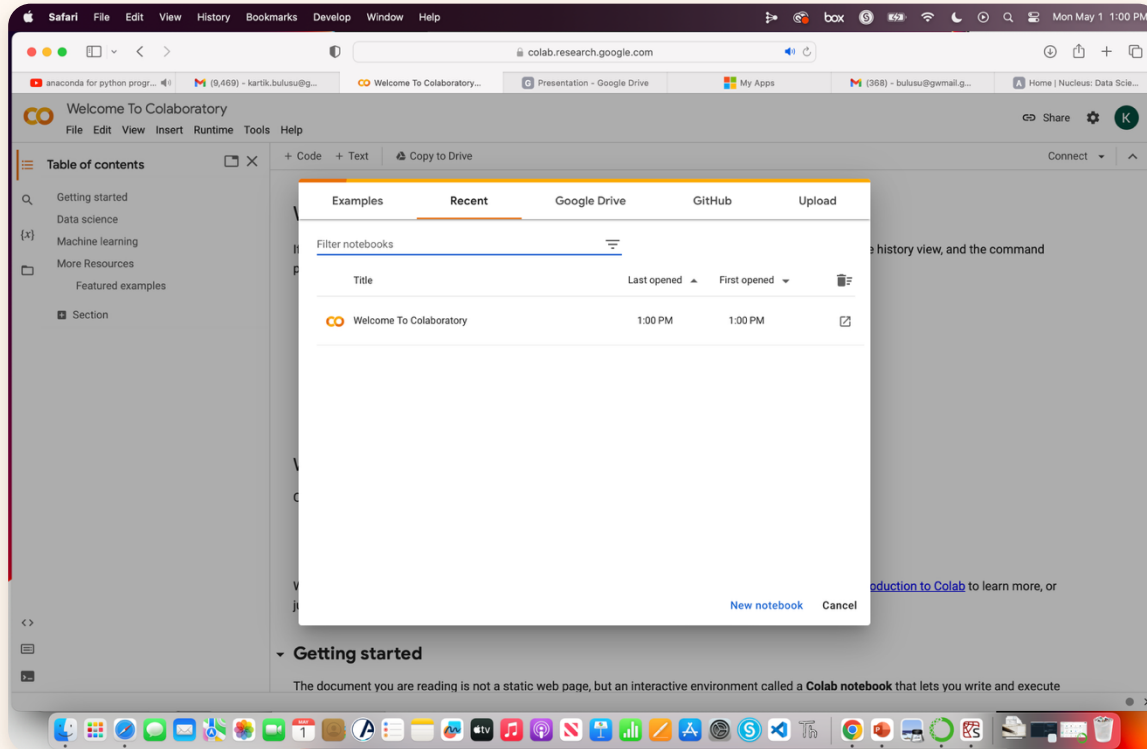
Google colab

Sources:

Google Colab <https://colab.research.google.com/>

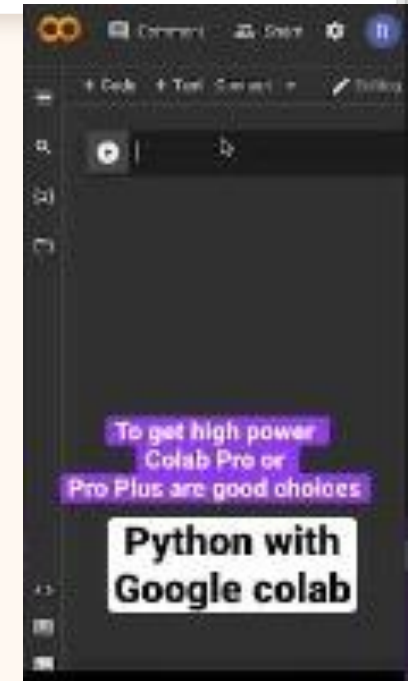
Google Workspace Marketplace <https://workspace.google.com/>

Youtube video shorts <https://youtube.com/shorts/7J3gU34j7Zk?feature=share>



Colaboratory ("Colab" for short)

- Data analysis and machine learning tool that allows you to
 - combine executable Python code and
 - rich text along with charts, images, HTML, LaTeX and
 - more into a single document stored in Google Drive.



It connects to powerful Google Cloud Platform runtimes and enables you to easily share your work and collaborate with others.



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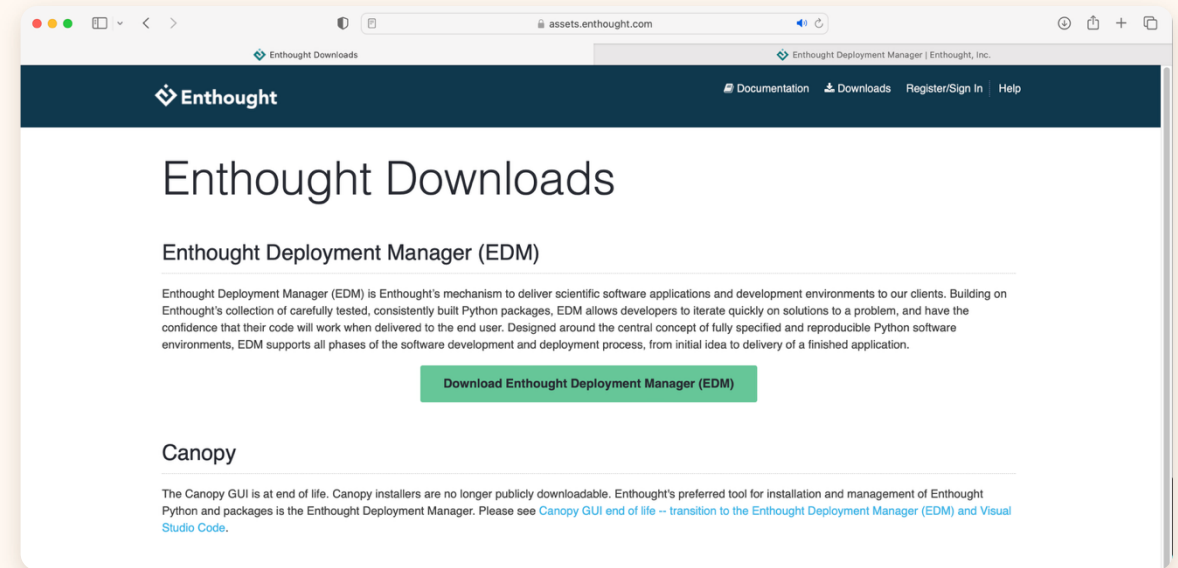
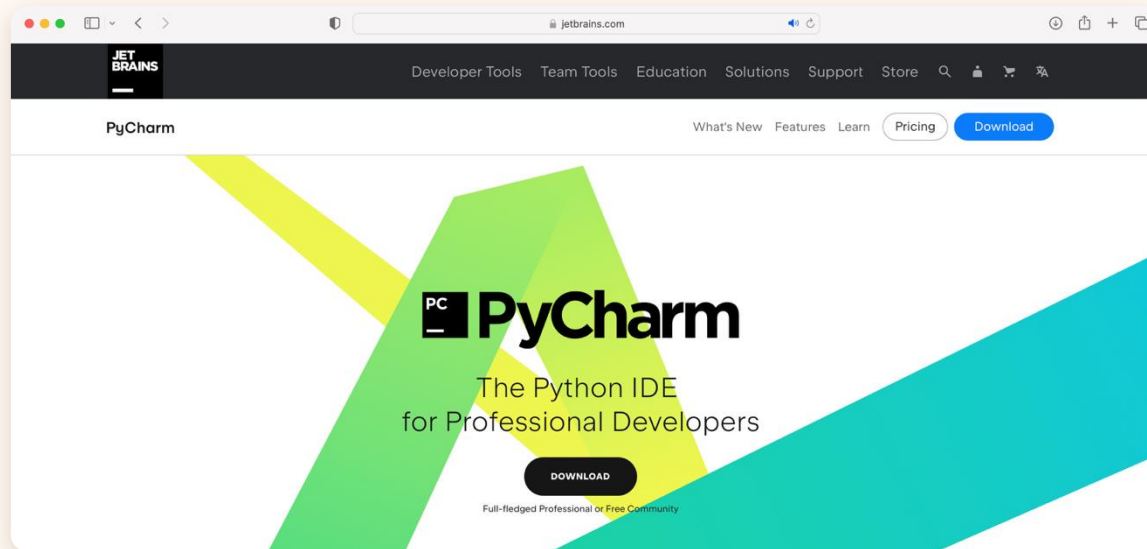


More...

Sources:

Jetbrains-Pycharm: <https://www.jetbrains.com/pycharm/>

Enthought Deployment Manager (EDM): <https://assets.enthought.com/downloads/>



Raffle with Python

